

Anytime-valid monitoring of repeated survey estimates under questionnaire redesign: An application to the Business Trends and Outlook Survey

Author information removed for anonymous peer review

Abstract

Statistical agencies release survey estimates every few weeks, and users read each release as it arrives, asking whether growth has changed course. Repeated significance testing on such a stream does not control error rates, and questionnaire redesigns create discontinuities that are easily mistaken for real change. We propose a monitor for the curvature of a published series. The null hypothesis is that the latent path is locally linear with unknown level and slope, with a separate unknown level for each questionnaire regime. A mixture e-process for the quadratic term, built on the residuals after removing these nuisance parameters, is valid under continuous inspection and optional stopping, provided that the published standard errors are correct, that redesign shifts are additive, and that the regime dates are known. In simulations with the sampling errors of the survey, the monitor holds its level at about 0.02, loses power to horizon-calibrated rules at small effects, and fails under proportional redesign effects and misdated regimes. Applied to 56 waves of the U.S. Census Bureau’s Business Trends and Outlook Survey on artificial intelligence use, it declares a deceleration on 10 March 2024, triggered by a single large drop, and an acceleration on 6 April 2025; the dates are stable to standard errors inflated by 55 percent, the evidence is not.

Keywords

sequential monitoring, e-values, survey redesign, repeated surveys, official statistics

1 Introduction

National statistical offices increasingly publish short-period estimates from rolling surveys. The U.S. Census Bureau’s Business Trends and Outlook Survey (BTOS) releases national, sector, state and size-class estimates every two weeks (U.S. Census Bureau 2025). Users, from journalists to central bank staff, read each release against its predecessors and ask whether the path of an indicator has changed, for example whether the adoption of a new technology has accelerated or stalled (Bonney et al. 2024; Federal Reserve Board 2026; McElheran et al. 2024). Two features of this practice are not handled by standard inferential tools.

The first is continuous inspection. A user who applies a conventional test at every release, and who may stop at any time, faces an error rate that is not the nominal one (Armitage et al. 1969). The second is redesign. Agencies revise questions, weighting and frames, and the published series then contains a discontinuity that is not a feature of the population. The survey literature handles this with models for repeated surveys that include an intervention or a parallel run, estimated by state-space methods that can be run in real time (Scott and Smith 1974; van den Brakel and Roels 2010; O’Hara et al. 2019; Durbin and Koopman 2012). These methods estimate the size of the discontinuity and of the underlying slope, which is what an analyst usually wants, but they do not give finite-sample error control for a user who inspects the series after every release.

Sequential analysis (Wald 1945; Page 1954; Brown et al. 1975; Lan and DeMets 1983; Chu et al. 1996) and its modern form based on e-values and test martingales (Howard et al. 2021; Vovk and Wang 2021; Ramdas et al. 2023; Grünwald et al. 2024; Shin et al. 2023), including always-valid monitoring of online experiments (Johari et al. 2022), provide tests with error control that is uniform over time. We combine this with the removal of nuisance parameters by projection onto the residual space, in the spirit of invariance arguments for e-values (Pérez-Ortiz et al. 2024). The construction is a Robbins mixture martingale for one coefficient of a Gaussian regression with known variances; its novelty lies in the choice of the null (local linearity with a separate level for each questionnaire regime), in the treatment of regimes that enter the design sequentially, and in the evaluation on a real survey series, not in the martingale itself.

The contributions are as follows. First, a monitor for curvature of a published survey series with a closed form, a proof of validity for all values of the level, the slope and the regime offsets (Section 3). Second, a simulation study that reports where the monitor works and where it does not: it holds its level, it is less powerful than horizon-calibrated rules at small effects, and it fails when the redesign effect is proportional, when the regime date is wrong, or when the published standard errors are too small (Section 4). Third, an application to 56 BTOS waves in which we report the sensitivity of each declaration to the prior scale, to inflated standard errors, to a tolerance for small curvature, and to a single large drop that dominates the first declaration (Section 5).

2 Setting

Let $j = 1, 2, \dots$ index published waves, with release times t_j measured in two-week units and published estimates y_j with design-based standard errors s_j . Each wave carries a questionnaire regime label $r_j \in \{1, \dots, K\}$ that is fixed by the agency

before the estimate is released. We write $w_j = 1/s_j^2$. The model for the published estimate is

$$y_j = \theta(t_j) + \mu_{r_j} + \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, s_j^2) \text{ independently,} \quad (1)$$

where θ is a smooth latent path and μ_r is the offset of regime r relative to the first regime. A redesign changes μ_r and leaves θ unchanged. The model is additive on the scale of the estimate: a redesign that rescales the series, or that changes its dynamics, is outside it. The standard errors s_j are treated as known and as fixed in advance, as is customary in the analysis of published time series; Subsections 4.1 and 5.2 examine what happens when they are too small. The normal approximation is reasonable at the sample sizes involved, and consecutive waves are drawn from different rotation panels (U.S. Census Bureau 2025), although a panel returns after twelve weeks and rotation group effects are a known source of bias in panel surveys (Bailar 1975).

The question asked of the series is about shape: over a stretch of consecutive waves, is the growth rate constant, or has it changed? For an epoch beginning at wave j_0 the null hypothesis is

$$H_0 : \quad \theta(t) = a + bt \quad \text{for some } a, b \in \mathbb{R}, \quad (2)$$

with a, b and the regime offsets μ_r unknown. The alternative is constant curvature, $\theta(t) = a + bt + \gamma t^2/2$, with $\gamma > 0$ for acceleration and $\gamma < 0$ for deceleration. A sustained change of the growth rate in one direction has a nonzero projection on the quadratic term and is detected as a departure from H_0 . Exact linearity is a null that no smooth path satisfies, so Subsection 3.1 also gives a version with a tolerance for small curvature.

3 Method

3.1 A mixture e-process for curvature

Fix an epoch and consider its first n waves. Let D_n be the $n \times p$ matrix whose columns are the indicators of the regimes present and the time variable t_j , let $W_n = \text{diag}(w_1, \dots, w_n)$, and let $q_j = t_j^2/2$. Define $M_n = I - D_n(D_n^\top W_n D_n)^{-1} D_n^\top W_n$ and

$$A_n = q^\top W_n M_n y, \quad B_n = q^\top W_n M_n q, \quad (3)$$

with q and y truncated to the first n waves. Here B_n is the information about γ after the level of each regime and the common slope have been removed by weighted least squares, and A_n/B_n is the corresponding estimate of γ , with standard error $B_n^{-1/2}$. Neither depends on how t_j is centred.

For a prior scale $\tau > 0$ on γ (in percentage points per squared wave) and a tolerance $\gamma_0 \geq 0$, the one-sided half-normal mixtures are

$$E_n^+ = \frac{2}{\tau \sqrt{a_n}} \Phi\left(\frac{A_n^+}{\sqrt{a_n}}\right) \exp\left(\frac{(A_n^+)^2}{2a_n}\right), \quad E_n^- = \frac{2}{\tau \sqrt{a_n}} \Phi\left(\frac{A_n^-}{\sqrt{a_n}}\right) \exp\left(\frac{(A_n^-)^2}{2a_n}\right), \quad (4)$$

where $a_n = B_n + \tau^{-2}$, $A_n^+ = A_n - \gamma_0 B_n$, $A_n^- = -A_n - \gamma_0 B_n$ and Φ is the standard normal distribution function. With $\gamma_0 = 0$ this is the closed form of $\int_0^\infty \exp(\gamma A_n - \gamma^2 B_n/2) \pi_\tau(d\gamma)$, with π_τ the half-normal law (Robbins 1970); for $\gamma_0 > 0$ the likelihood ratio is taken against the boundary value γ_0 , and the formula is the same with A_n shifted by $\gamma_0 B_n$. The statistic E_n^+ is evidence for acceleration and E_n^- for deceleration. Monitoring starts at the first wave n with $B_n > 0$, which is the third wave of an epoch whose waves share a regime.

Proposition 1. *Let $\alpha \in (0, 1)$. Suppose model (1) holds with t_j , s_j and r_j known in advance, and $\gamma \leq \gamma_0$ (that is, H_0 when $\gamma_0 = 0$). Then for every a, b and regime offsets μ_r ,*

$$\mathbb{P}(\exists n : E_n^+ \geq 2/\alpha) \leq \alpha/2,$$

and likewise for E^- when $\gamma \geq -\gamma_0$. The probability that either monitor ever reaches $2/\alpha$ is therefore at most α .

The proof is in Appendix A. The guarantee holds at all times simultaneously, so the user may inspect the series after every release and stop at any time. The declaration rule is the first time either monitor reaches $2/\alpha$ (40 for $\alpha = 0.05$). Below the threshold the e-values are not given verbal labels, because under the null the probability that E^+ or E^- ever exceeds a moderate level is not small: over 52 looks it is about 0.58 for a level of 2 and 0.09 for 10. As a summary of the evidence in an epoch without a declaration we report the anytime-valid p-value $\min\{1, 2/\max_n \max(E_n^+, E_n^-)\}$.

3.2 Redesigns as nuisance levels

The regime offsets μ_r enter D_n as separate columns. A new regime adds a column that is zero on all earlier rows, so the residual σ -fields used in the proof remain nested. Proposition 1 then holds for every value of the offsets, and a redesign that shifts the level by a constant of any size cannot produce a false declaration. Three conditions are needed and none is innocuous. The shift must be additive on the scale of the series; the post-redesign slope and curvature must be the same as before the redesign; and the regime label must be placed at the correct wave. Subsection 4.3 shows that a proportional effect, or a label one wave late, destroys the guarantee. The size of the jump is not used, which costs information: a regime with

Table 1. Probability of at least one declaration over 52 looks (20,000 replications; nominal level 0.05). The first three rows are valid cases; correlated errors use a lag-6 autoregressive structure at the stated correlation, and standard errors are too small by the stated factor. Rows 4 to 10 use slope 0.15.

Scenario	E-process	Bonferroni	Spending	Exact boundary	Uncorrected
Linear path, slope 0.00	0.019	0.023	0.042	0.048	0.601
Linear path, slope 0.15	0.018	0.025	0.046	0.049	0.602
Linear path, slope 0.30	0.020	0.025	0.045	0.049	0.597
Lag-6 correlation 0.25	0.011	0.022	0.043	0.044	0.573
Lag-6 correlation 0.50	0.006	0.019	0.043	0.037	0.498
Lag-6 correlation 0.80	0.000	0.010	0.045	0.019	0.323
True SE 1.10 times published	0.047	0.062	0.086	0.109	0.733
True SE 1.25 times published	0.119	0.160	0.177	0.243	0.876
True SE 1.50 times published	0.318	0.415	0.398	0.524	0.971
True SE 1.75 times published	0.531	0.661	0.628	0.756	0.994

one wave is absorbed completely, and a regime with two waves contributes one within-regime difference, so over a short post-redesign run the regime-aware monitor is equivalent to dropping the transition wave.

For comparison we also report the pooled analysis, in which the regime columns are dropped and the spliced series is treated as one regime.

3.3 Epochs and restarts

After a declaration the monitor starts a new epoch at the following wave, with fresh level and slope parameters.

Corollary 1. *Suppose the latent path is linear on the epoch's span so far. For an epoch beginning at a stopping time, the conditional probability that it ends in a declaration is at most α . If the latent path is linear throughout, the number N of declarations satisfies $\mathbb{P}(N \geq m) \leq \alpha^m$ and $\mathbb{E}[N] \leq \alpha/(1 - \alpha)$.*

This is the sense of error control for a sequence of declarations. It says that a declaration is rarely spurious when the path is linear. It says nothing when the path is not linear, which for any smooth adoption curve is the typical case: a declaration then records that the epoch's path has curved, and with the information B_n growing, small curvature is eventually declared. The tolerance γ_0 addresses this: with $\gamma_0 > 0$ the guarantee holds for every path whose curvature stays below γ_0 . A declaration is also a crossing time, not an estimate of when the change occurred, and the dates depend on where the epoch starts. We therefore report the estimated curvature A_n/B_n with its standard error at each declaration.

3.4 Cross-strata attribution check

When an agency announces a redesign the regime label is known. When it does not, the label may have to be inferred. Let $z_{k,j}$ be the standardized change in stratum k between waves $j - 1$ and j . An instrument event moves every stratum in the same wave, with the same sign and with magnitudes far above the stratum's own distribution of within-regime movements, whereas a change in the latent path builds up over several waves. We compare the change of each stratum with the 95th percentile of its own within-regime $|z_{k,j}|$, and count the strata beyond it with the same sign. This is a screening rule with no formal error control. We apply it to every wave and not only to announced redesigns.

3.5 Constants

The level is $\alpha = 0.05$, and each monitor is run at threshold $2/\alpha = 40$. The prior scale $\tau = 0.01$ percentage points per squared wave is a change of 0.1 points per wave in the growth rate accumulating over ten waves. This scale is not derived from a power calculation; the value was used in all analyses reported here, but other values were not excluded in advance, and the results for τ between 0.0025 and 0.04 are reported in Subsection 5.4. Validity does not depend on τ , only power and delay do. The tolerance is $\gamma_0 = 0$ unless stated.

4 Simulation study

All simulations use the dates and design-based standard errors of the 54 legacy-wording BTOS waves, with 20,000 replications unless stated. Five rules monitor the curvature statistic $A_n/\sqrt{B_n}$ over the 52 looks from the third wave. The e-process uses $\tau = 0.01$ and $\alpha = 0.05$. The Bonferroni rule uses level $\alpha/52$ at each look and needs the horizon in advance. The spending rule uses level $\alpha/(k(k+1))$ at look k and does not. The exact boundary is a constant threshold on $|A_n|/\sqrt{B_n}$, calibrated by simulation to size 0.05 for this design; it needs the horizon and the information schedule. The uncorrected rule tests at level α at every look. The simulation code is part of the replication package.

Table 2. Takeoff detection when the growth rate rises by the stated amount (percentage points per wave) at the 25th of 54 waves, from a baseline growth of 0.10. Each cell gives the probability of a first declaration within 20 waves of the change, the probability of one by the end of the series, and the median delay in waves among the paths detected. The probability of a declaration before the change is 0.012 to 0.015 for the e-process and the Bonferroni rule, 0.04 for the spending rule and 0.03 for the exact boundary.

Slope increase	E-process	Bonferroni	Spending	Exact boundary
0.01	0.03 / 0.08 / 22	0.06 / 0.15 / 23	0.01 / 0.02 / 24	0.09 / 0.21 / 22
0.02	0.29 / 0.63 / 21	0.41 / 0.78 / 20	0.12 / 0.39 / 24	0.49 / 0.83 / 19
0.03	0.80 / 0.97 / 16	0.87 / 0.98 / 16	0.58 / 0.91 / 19	0.89 / 0.97 / 14
0.05	0.99 / 0.99 / 11	0.99 / 0.99 / 10	0.96 / 0.96 / 13	0.97 / 0.97 / 10

Table 3. Probability of a false declaration after a redesign at the 21st wave, with 2, 5, 10 or 20 waves in the new regime, on a linear path of slope 0.25. The additive shift is 7.3 points; the multiplicative effect scales the series by 1.73; in the third column the shift is additive but the regime label is placed one wave late. The last column is the pooled analysis with an additive shift. 5,000 replications.

Waves after redesign	Additive shift	Multiplicative effect	Label one wave late	Pooled analysis
2	0.012	0.014	1.000	1.000
5	0.015	0.028	1.000	1.000
10	0.016	0.999	1.000	1.000
20	0.019	1.000	1.000	1.000

4.1 False-declaration probability

Table 1 shows that the e-process holds its level for all three slopes at about 0.02, below the nominal 0.05, because the mixture and the two-sided split are conservative at this horizon. The Bonferroni rule is similar at 0.023 to 0.025, the spending rule uses most of the budget (0.043 to 0.046), the exact boundary reaches its size of 0.05 by construction, and the uncorrected rule declares at least once in 60 percent of replications, 2.6 times on average. Over the full restart procedure under a linear path the probability of at least one declaration is 0.022 and the expected number of declarations is 0.022 (slope 0) and 0.024 (slope 0.3), within the bound of Corollary 1. Lag-6 correlation, as when a rotating panel returns, makes the e-process, the Bonferroni rule and the exact boundary more conservative over these 52 looks and leaves the spending rule unchanged; this direction is not general and may reverse for longer epochs. The weak point is the standard errors. If the true standard error is 1.25 times the published one, the e-process declares falsely in 12 percent of replications, with 1.5 times in 32 percent and with 1.75 times in 53 percent, and the other rules are no better. The e-process is not more robust than the others; it is only less sensitive because it is more conservative.

With a tolerance $\gamma_0 = 0.001$, a path with curvature $\gamma = 0.001$ is declared by the monitor with $\gamma_0 = 0$ in 97 percent of replications and by the monitor with the tolerance in 1.1 percent; with $\gamma = 0.003$ both declare in all replications.

4.2 Detection of a takeoff

Table 2 shows that the e-process is less powerful than the Bonferroni rule and the exact boundary at small effects: for a slope increase of 0.02 the probability of a declaration by the end is 0.63, against 0.78 and 0.83, and the median delay is 21 waves against 20 and 19. At larger effects the three are close (0.97 to 0.99 by the end for a slope increase of 0.05, with a median delay of 11, 10 and 10 waves). The spending rule, which like the e-process does not need the horizon, is less powerful than both at every effect size. The cost of the e-process relative to the horizon-calibrated rules is therefore a loss of power at small effects, partly from its realized size of 0.02 and partly from the prior scale, in exchange for a guarantee that does not depend on the horizon.

4.3 Redesigns

Table 3 separates what the guarantee covers from what it does not. For an additive shift the regime-aware monitor holds its level at every length of the new regime, and the pooled analysis declares in every replication. For a proportional effect, which scales the post-redesign path and so changes its slope, the monitor is fine for 2 and 5 new waves (0.014 and 0.028), because the new regime carries little information, and fails from 10 waves on (0.999 at 10). A label placed one wave late fails at every length. A short post-redesign run, as in the BTOS application, therefore cannot reveal whether the redesign effect is additive.

5 Application: business use of artificial intelligence

5.1 Data

The BTOS is an experimental survey of about 1.2 million nonfarm employer businesses divided into six rotating panels; each panel reports every twelve weeks and collection occurs every two weeks (U.S. Census Bureau 2025). Two questions on artificial intelligence have been asked since the collection period beginning 11 September 2023 (Bonney et al. 2024). Until the collection period ending in October 2025 the question on current use read “in producing goods or services”; from the collection beginning 17 November 2025 it read “in any of its business functions” (U.S. Census Bureau 2026). The 15

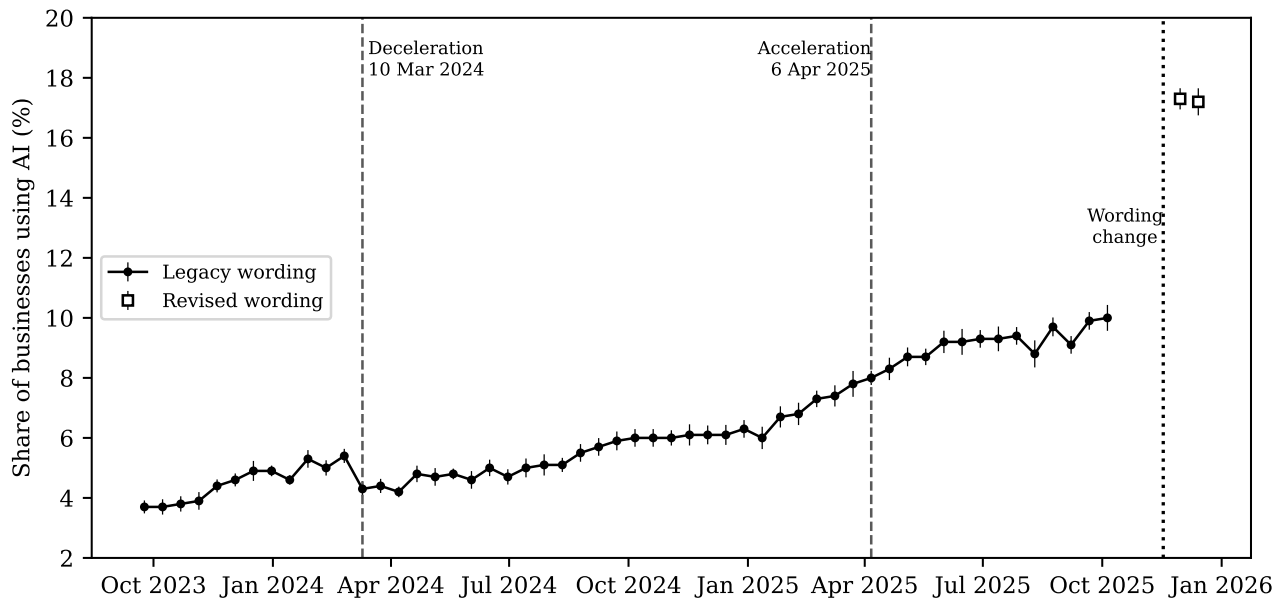


Figure 1. National estimate of the share of businesses using artificial intelligence, September 2023 to December 2025, with 95 percent intervals. Dashed vertical lines mark the two declarations. The dotted vertical line marks the wording change of 17 November 2025. Circles: legacy wording; open squares: revised wording.

Table 4. Epochs of the regime-aware monitor on the national series ($\tau = 0.01$, $\alpha = 0.05$). Curvature is the estimate A_n/B_n at declaration with a 95 percent half-width, conditional on the published standard errors. Peak E is the value at declaration, or the maximum over the epoch when it ends without one; for the last epoch the anytime-valid p-value is 0.105.

Epoch start	Waves	Declaration	Date	Curvature (pp per wave ²)	Peak E
24 Sep 2023	13	Deceleration	10 Mar 2024	-0.0281 ± 0.0101	1.0×10^5
24 Mar 2024	28	Acceleration	6 Apr 2025	0.0047 ± 0.0017	1.9×10^5
20 Apr 2025	15	None	—	—	19.0

December 2025 file of the Census Bureau removed the legacy-wording rows, so we combine the archived vintages of 8 and 15 December 2025 (Goldschlag 2025). The data set has 56 published national waves: 54 legacy-wording waves with collection ending 24 September 2023 to 5 October 2025, and two revised-wording waves ending 30 November and 14 December 2025. No estimate was published for the three intervening periods, which appear as a gap in the time index. National standard errors have a median of 0.15 points (range 0.08 to 0.23). Estimates and standard errors are also available for seven employment size classes. The series ends in December 2025. The data are an archived final series and not the vintages that a live user would have seen, so the dates below describe the series as it now stands.

Under the legacy wording the national estimate rose from 3.7 percent to 4.9 percent by the end of 2023, fell to 4.3 percent in March 2024, and then rose to 10.0 percent in October 2025. Under the revised wording it was 17.3 and 17.2 percent.

5.2 Dispersion of the published standard errors

The standardized wave-to-wave changes within the legacy regime have a lag-one autocorrelation of -0.49 , as expected for independent errors around a slowly moving path. The standard errors themselves are the weak point. The standardized second differences, which remove a local linear trend, should have a standard deviation near one if the path is smooth and the errors are as published; they have 1.76 (1.55 by a robust estimate based on the median absolute deviation). Some of this is genuine roughness of the latent path, some is understated standard errors, and the two cannot be separated from the published figures alone. By Table 1 a factor of 1.5 on the standard errors is in the range where false declarations are frequent, so we report the results for inflated standard errors in Subsection 5.4.

5.3 Declarations

Figure 1 and Table 4 give the results. The first epoch begins on 24 September 2023 and the deceleration monitor declares on 10 March 2024, after 13 waves, with an estimated curvature of -0.028 ± 0.010 points per squared wave. The declaration is triggered by a single wave: the estimate falls from 5.4 to 4.3 percent, a change of -6.2 combined standard errors, the largest within-regime change in the series, and the e-value moves from 1.2 to 1.0×10^5 in that wave. The estimate stays near 4.4 afterwards. The cross-strata check finds three of the seven size classes beyond their own 95th percentile with the same sign at that wave, which is the only legacy wave with three or more, against seven of seven at the wording change. The evidence therefore does not distinguish a genuine fall in business use from an unannounced collection or processing change. If the

Table 5. Declarations with the published standard errors multiplied by the stated factor (1.55 is the robust estimate from the second differences).

SE inflation	First declaration (E)	Second declaration (E)	Peak E, third epoch
1.00	10 Mar 2024 (1.0×10^5)	6 Apr 2025 (1.9×10^5)	19.0
1.25	10 Mar 2024 (834.9)	6 Apr 2025 (1.5×10^3)	5.3
1.50	10 Mar 2024 (72.7)	6 Apr 2025 (117.5)	2.9
1.55	10 Mar 2024 (52.0)	6 Apr 2025 (81.8)	2.7

Table 6. Employment size classes at the wording change. Ratio is the first revised estimate over the last legacy estimate; jump is the standardized change; the 95th percentile is that of the class's standardized wave-to-wave changes within the legacy regime. The last column gives the date on which the pooled monitor declares an acceleration, where it does.

Size class	Last legacy	First revised	Ratio	Jump $ z $	95th pct. within	Pooled declares
1–4	11.0	17.0	1.55	13.2	2.77	30 Nov 2025
5–9	8.6	15.5	1.80	8.7	2.85	30 Nov 2025
10–19	8.1	17.8	2.20	10.6	2.37	30 Nov 2025
20–49	7.0	18.7	2.67	13.8	1.96	30 Nov 2025
50–99	9.7	20.7	2.13	9.7	2.29	30 Nov 2025
100–249	8.8	23.7	2.69	8.6	1.74	no
250+	13.0	30.5	2.35	6.1	2.31	no

wave is dropped, the first declaration is a deceleration on 24 March 2024 and the second is unchanged; if the wave is treated as the start of a new level that persists, there is no first declaration and the sequence becomes three decelerations (30 June 2024, 12 January 2025, 27 July 2025) with no acceleration. The first declaration should be read as conditional on the March 2024 wave being a feature of the population.

The second epoch begins on 24 March 2024 and the acceleration monitor declares on 6 April 2025, after 28 waves, with an estimated curvature of 0.0047 ± 0.0017 : the growth rate was rising by about 0.1 points per wave over 20 waves. The third epoch begins on 20 April 2025 and runs through the wording change; the deceleration monitor, which responds to the flattening of the series through August 2025, peaks at 19.0 on 7 September 2025, an anytime-valid p-value of 0.105, and does not declare. The acceleration monitor stays below 1.1. The two revised-wording waves are retained with a level of their own.

5.4 Sensitivity

The two declarations are stable to the prior scale: for τ from 0.005 to 0.04 they occur on the same two dates, and for $\tau = 0.0025$ the first is two weeks later (24 March 2024). They are stable in date to inflated standard errors up to the estimated factor of 1.55 (Table 5), but not in strength: the e-values fall from about 10^5 to 52 and 82, barely above the threshold. They are stable to a tolerance of $\gamma_0 = 0.001$ (e-values 4.5×10^4 and 917); with $\gamma_0 = 0.002$ the second moves to 20 April 2025 (e-value 194). The third epoch is not stable. With $\tau = 0.02$ the deceleration is declared on 7 September 2025 ($E = 89$) and with $\tau = 0.04$ on 10 August 2025 ($E = 45$); at $\alpha = 0.10$ it is declared on 10 August 2025; with standard errors inflated by 1.25 the peak falls to 5.3. Replacing the published standard errors by their median of 0.15 gives a different sequence (10 March 2024, 23 March 2025, 10 August 2025), so the outcome of the third epoch depends on the weights given to individual waves. The summer 2025 slowdown is best described as inconclusive.

5.5 The wording change

Between the last legacy wave (10.0 percent, 5 October 2025) and the first revised wave (17.3 percent, 30 November 2025) the estimate rose by 7.3 points, which is 25.7 combined standard errors, against a within-regime median absolute standardized change of 0.94 and a maximum of 6.2. The regime-aware monitor books no evidence from it, by construction, and with two waves in the new regime this says little beyond that the transition wave is absorbed. The pooled analysis declares an acceleration on 30 November 2025 with an E-value of order 10^{123} , and a repeated curvature test applied to the spliced series has a statistic of 36.4 in magnitude at that wave.

Table 6 shows that all seven size classes jump in the same wave and in the same direction, by 6.1 to 13.8 standard errors and by 2.6 to 7.0 times their own 95th percentile (1.74 to 2.85). The ratio of the first revised to the last legacy estimate, 1.55 to 2.69, rises with the level of the class, so the effect on the scale of the series is closer to proportional than to a constant; at the national level the ratio is 1.73. With two revised waves we cannot test whether the effect is additive, and Table 3 shows that this matters once the new regime is longer.

5.6 Comparison with repeated testing

The curvature z statistic was applied at each wave to the series with the same restart rule, with regime columns and without. The uncorrected test declares nine times, the first on 19 November 2023. The Bonferroni rule, using the remaining length of the series as its horizon, declares on 10 March 2024, 6 April 2025, 10 August 2025 and 21 September 2025; the last two are

the summer 2025 slowdown, which the e-process at $\tau = 0.01$ does not declare, as expected from the loss of power in Table 2. The spending rule declares six times, with the first and the last two on the same dates as the Bonferroni rule. A moving-average rule, which crosses when the sign of the 3-wave mean minus the 8-wave mean changes, crosses four times (24 March 2024, 19 May 2024, 7 September 2025, 21 September 2025). Without regime columns the Bonferroni and spending rules add a declaration on 14 December 2025 and the uncorrected rule declares on 30 November 2025, at the revised-wording waves. With regime columns the first two dates agree with the e-process. The e-process is therefore not distinguished on the real data by what it finds; its advantage is the guarantee, at a cost in power.

6 Discussion

6.1 Implications for agencies and users

Agencies that announce a redesign should publish the regime label with the series; the monitor needs it, and an error of one wave is enough to destroy the guarantee. Users who monitor a series should report the epoch, the estimated curvature and its standard error, and the sensitivity to the standard errors, in addition to the e-value. The e-process is a complement to, not a substitute for, a state-space model with an intervention, which estimates the slope and its change directly; it is useful where a user wants a stopping rule with a stated error probability that holds however often the series is inspected.

6.2 Limitations

The treatment of the published standard errors as exact is the principal limitation: Table 1 shows false declarations of 12 percent when they are 25 percent too small, and the BTOS series shows dispersion of the size at which this happens. The standard errors are estimates, not fixed constants, which the proof does not cover; a time-uniform treatment of unknown variance exists for simpler models (Wang and Ramdas 2025). Redesign effects must be additive and the regime dates exact. The null is a statement about shape, and a declaration says that the path has curved, not why; the first declaration in the application is driven by one large drop that may be a measurement event, and the cross-strata check is a screening rule. Declarations depend on where the epoch starts, which is a modelling choice, and the dates are crossing times. The data are an archived final series ending in December 2025, and the post-redesign run is two waves long. Nonresponse bias that drifts over time is absorbed into the latent path and reported as population change.

7 Conclusion

A published survey series can be monitored for changes in its growth rate with error control that holds under continuous inspection, for any unknown level and slope and for additive redesign shifts at known dates, provided that the published standard errors are right. Applied to the BTOS series on business use of artificial intelligence, the monitor declares a deceleration in March 2024, triggered by one large drop, and an acceleration in April 2025; it leaves the summer 2025 slowdown inconclusive. The guarantee fails for proportional redesign effects, misdated regimes and understated standard errors, and the application cannot rule these out.

A Proof of Proposition 1 and Corollary 1

Fix an epoch and write $\mathcal{X}_n = \text{span}(D_n) \subset \mathbb{R}^n$. Under model (1) the vector $y_{1:n}$ equals $D_n c + \gamma q_{1:n} + e_{1:n}$ for some $c \in \mathbb{R}^p$ and $e_{1:n} \sim \mathcal{N}(0, W_n^{-1})$. Let G_n be the σ -field generated by $R_n = M_n y_{1:n}$, that is, by $y_{1:n}$ modulo $\mathcal{X}_n$. The columns of D_{n+1} restricted to the first n rows span $\mathcal{X}_n$, with a new regime column being zero on earlier rows, so $y_{1:n+1} + \mathcal{X}_n$ is a function of $y_{1:n+1} + \mathcal{X}_{n+1}$ and (G_n) is a filtration. Moreover $B_{n+1} \geq B_n$: the residual of q on the larger design restricted to the first n rows is no larger in the W -norm than the residual on D_n , and the new row adds a nonnegative term.

In the inner product $\langle u, v \rangle_W = u^\top W_n v$, M_n is the orthogonal projector onto $\mathcal{X}_n^\perp$. The vector R_n is a standard Gaussian on $\mathcal{X}_n^\perp$ with mean $\gamma M_n q$, and its law does not depend on c . Write $u = \delta - \gamma_0$ for a candidate curvature $\delta > \gamma_0$ and $A_n^+ = A_n - \gamma_0 B_n$, where $\langle M_n q, R_n \rangle_W = A_n$ and $\langle M_n q, M_n q \rangle_W = B_n$. The density ratio of R_n under curvature δ against curvature γ_0 is

$$L_n(u) = \exp(u A_n^+ - \frac{1}{2} u^2 B_n).$$

For $\gamma = \gamma_0$, (L_n) is the likelihood ratio of the observation R_n of fixed law on the increasing sequence G_n and is a nonnegative martingale with initial value one. Mixing over $u > 0$ with the half-normal law keeps this and gives E_n^+ by evaluating the Gaussian integral,

$$\int_0^\infty e^{u A_n^+ - \frac{1}{2} u^2 B_n} \frac{2}{\tau \sqrt{2\pi}} e^{-u^2/(2\tau^2)} du = \frac{2}{\tau \sqrt{a}} \Phi\left(\frac{A}{\sqrt{a}}\right) e^{A^2/(2a)}, \quad a = B + \tau^{-2}.$$

For $\gamma < \gamma_0$, conditionally on G_{n-1} the increment $A_n^+ - A_{n-1}^+$ is normal with mean $(\gamma - \gamma_0)(B_n - B_{n-1})$ and variance $B_n - B_{n-1}$ (the innovation of a Gaussian regression in sequentially orthogonalized form). Hence $\mathbb{E}[L_n/L_{n-1} \mid G_{n-1}] = \exp(u(\gamma - \gamma_0)(B_n - B_{n-1})) \leq 1$ for $u > 0$, because $B_n - B_{n-1} \geq 0$, and the mixture is a supermartingale. Ville's inequality gives $\mathbb{P}(\sup_n E_n^+ \geq 2/\alpha) \leq \alpha/2$ for all c . The case of E^- is the same with the sign of A_n reversed. This proves Proposition 1.

For Corollary 1, condition on the history up to the stopping time at which the epoch starts. Given the history, the future waves obey the model and the latent path is a fixed function. If the path is linear on the first n^* waves of the epoch, a declaration at $n \leq n^*$ is an event of the type bounded in Proposition 1, applied to the model whose path is the linear extension, which has the same law of the data up to n^* . If the path is linear throughout, each epoch ends in a declaration with conditional probability at most α , so $\mathbb{P}(N \geq m) \leq \alpha^m$ and $\mathbb{E}[N] = \sum_{m \geq 1} \mathbb{P}(N \geq m) \leq \alpha/(1 - \alpha)$.

B Computation

For each epoch, at each wave n with $B_n > 0$ the weighted regression of y and q on regime indicators and t gives the residuals; A_n and B_n follow from (3), and $E_n^\pm$ is evaluated on the logarithmic scale. If $\max(E_n^+, E_n^-) \geq 40$ a declaration is recorded and the next epoch starts at wave $n + 1$. All results were produced by the scripts in the replication package, from the data of Subsection 5.1.

Declaration of Conflicting Interests

The Author declares that there is no conflict of interest.

Funding

The Author received no financial support for the research, authorship, and/or publication of this article.

Data Availability

All data are public statistics of the U.S. Census Bureau (Business Trends and Outlook Survey, https://www.census.gov/hfp/btos/data_downloads). The analysis data set and the code that reproduces every number, table and figure are provided with the submission.

References

- Armitage, P., C. K. McPherson, and B. C. Rowe (1969). Repeated significance tests on accumulating data. *Journal of the Royal Statistical Society: Series A* 132(2), 235–244. <https://doi.org/10.2307/2343787>.
- Bailar, B. A. (1975). The effects of rotation group bias on estimates from panel surveys. *Journal of the American Statistical Association* 70(349), 23–30. <https://doi.org/10.1080/01621459.1975.10480255>.
- Bonney, K., C. Breaux, C. Buffington, E. Dinlersoz, L. Foster, N. Goldschlag, J. Haltiwanger, Z. Kroff, and K. Savage (2024). Tracking firm use of AI in real time: A snapshot from the Business Trends and Outlook Survey. NBER Working Paper 32319, National Bureau of Economic Research. <https://doi.org/10.3386/w32319>.
- Brown, R. L., J. Durbin, and J. M. Evans (1975). Techniques for testing the constancy of regression relationships over time. *Journal of the Royal Statistical Society: Series B* 37(2), 149–163. <https://doi.org/10.1111/j.2517-6161.1975.tb01532.x>.
- Chu, C.-S. J., M. Stinchcombe, and H. White (1996). Monitoring structural change. *Econometrica* 64(5), 1045–1065. <https://doi.org/10.2307/2171955>.
- Durbin, J. and S. J. Koopman (2012). *Time Series Analysis by State Space Methods* (2nd ed.). Oxford: Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199641178.001.0001>.
- Federal Reserve Board (2026). Monitoring AI adoption in the U.S. economy. FEDS Notes, April 3, 2026. Board of Governors of the Federal Reserve System. <https://www.federalreserve.gov/econres/notes/feds-notes/monitoring-ai-adoption-in-the-u-s-economy-20260403.html>.
- Goldschlag, N. (2025). How many businesses are using AI? Agglomerations, December 19, 2025. <https://agglomerations.substack.com/p/how-many-businesses-are-using-ai>.
- Grünwald, P., R. de Heide, and W. M. Koolen (2024). Safe testing. *Journal of the Royal Statistical Society: Series B* 86(5), 1091–1128. <https://doi.org/10.1093/jrsssb/qkae011>.
- Howard, S. R., A. Ramdas, J. McAuliffe, and J. Sekhon (2021). Time-uniform, nonparametric, nonasymptotic confidence sequences. *Annals of Statistics* 49(2), 1055–1080. <https://doi.org/10.1214/20-AOS1991>.
- Johari, R., P. Koomen, L. Pekelis, and D. Walsh (2022). Always valid inference: Continuous monitoring of A/B tests. *Operations Research* 70(3), 1806–1821. <https://doi.org/10.1287/opre.2021.2135>.
- Lan, K. K. G. and D. L. DeMets (1983). Discrete sequential boundaries for clinical trials. *Biometrika* 70(3), 659–663. <https://doi.org/10.2307/2336502>.
- McElheran, K., J. F. Li, E. Brynjolfsson, Z. Kroff, E. Dinlersoz, L. Foster, and N. Zolas (2024). AI adoption in America: Who, what, and where. *Journal of Economics & Management Strategy* 33(2), 375–415. <https://doi.org/10.1111/jems.12576>.
- O'Hara, A., C. Medalia, and J. Maples (2019). Modeling a bridge when survey questions change: Evidence from the Current Population Survey Health Insurance redesign. *Journal of Official Statistics* 35(1), 189–202. <https://doi.org/10.2478/jos-2019-0009>.
- Page, E. S. (1954). Continuous inspection schemes. *Biometrika* 41(1–2), 100–115. <https://doi.org/10.1093/biomet/41.1-2.100>.

- Pérez-Ortiz, M. F., T. Lardy, R. de Heide, and P. Grünwald (2024). E-statistics, group invariance and anytime-valid testing. *Annals of Statistics* 52(4). <https://doi.org/10.1214/24-AOS2394>.
- Ramdas, A., P. Grünwald, V. Vovk, and G. Shafer (2023). Game-theoretic statistics and safe anytime-valid inference. *Statistical Science* 38(4), 576–601. <https://doi.org/10.1214/23-STS894>.
- Robbins, H. (1970). Statistical methods related to the law of the iterated logarithm. *Annals of Mathematical Statistics* 41(5), 1397–1409. <https://doi.org/10.1214/aoms/1177696786>.
- Scott, A. J. and T. M. F. Smith (1974). Analysis of repeated surveys using time series methods. *Journal of the American Statistical Association* 69(347), 674–678. <https://doi.org/10.1080/01621459.1974.10480187>.
- Shin, J., A. Ramdas, and A. Rinaldo (2023). E-detectors: A nonparametric framework for sequential change detection. *The New England Journal of Statistics in Data Science* 2(2), 229–260. <https://doi.org/10.51387/23-NEJSDS51>.
- U.S. Census Bureau (2025). Business Trends and Outlook Survey: Methodology. U.S. Census Bureau, Washington, DC. <https://www2.census.gov/data/experimental-data-products/business-trends-and-outlook-survey/methodology.pdf>.
- U.S. Census Bureau (2026). Business Trends and Outlook Survey data release, May 7, 2026. Press release. <https://www.census.gov/newsroom/press-releases/2026/btos-may-7.html>.
- van den Brakel, J. A. and J. Roels (2010). Intervention analysis with state-space models to estimate discontinuities due to a survey redesign. *Annals of Applied Statistics* 4(2). <https://doi.org/10.1214/09-AOAS305>.
- Vovk, V. and R. Wang (2021). E-values: Calibration, combination and applications. *Annals of Statistics* 49(3), 1736–1754. <https://doi.org/10.1214/20-AOS2020>.
- Wald, A. (1945). Sequential tests of statistical hypotheses. *Annals of Mathematical Statistics* 16(2), 117–186. <https://doi.org/10.1214/aoms/1177731118>.
- Wang, H. and A. Ramdas (2025). Anytime-valid t-tests and confidence sequences for Gaussian means with unknown variance. *Sequential Analysis* 44(1), 56–110. <https://doi.org/10.1080/07474946.2024.2428245>.